

Experiment 8: Statistical Analysis of Data using Sliding Window

Aim

To implement a sliding window algorithm that performs statistical analysis of a continuously arriving data stream, calculating the mean, median, variance, standard deviation, maximum, minimum and range for every window.

Statistical Analysis

Mean, median, variance, standard deviation, maximum, minimum and range.

Theory

In stream processing the data never ends, so statistics cannot be calculated on the complete dataset. Instead a window of fixed size is placed over the most recent elements of the stream. As each new element arrives the window moves forward by one position, dropping the oldest element and including the newest one. Statistics are then recalculated for the current window only. This gives an up to date picture of the recent behaviour of the stream while using a constant amount of memory, which is essential when the stream is very large or unbounded.

Algorithm

1. Read the streaming data.
2. Fix the window size.
3. Move the window as new data enters.
4. Extract the current window.
5. Perform statistical analysis.
6. Display the result.
7. Repeat the steps.

Total Number of Windows Created

$$\text{Total windows} = n - w + 1$$

where $n = 20$ and $w = 5$

$$\begin{aligned} &= 20 - 5 + 1 \\ &= 16 \end{aligned}$$

Requirements

- Python 3.x
- Jupyter Notebook
- Statistics module
- Random module

Step 1: Import Libraries

```
import statistics
import random
```

Interpretation: Imports the statistics module for the calculations and the random module for generating the stream.

Step 2: Read Streaming Data

```
stream = [random.randint(10, 100) for i in range(20)]
```

Interpretation: Generates a stream of 20 random values between 10 and 100.

Step 3: Fix the Window Size

```
window_size = 5  
print("Window size:", window_size)
```

Interpretation: Fixes the number of recent elements that will be analysed at a time.

Step 4: Move the Window as New Data Enters

```
for i in range(len(stream) - window_size + 1):  
    window = stream[i : i + window_size]
```

Interpretation: Slides the window forward one position at a time and extracts the current window from the stream.

Step 5: Perform Statistical Analysis

```
mean = statistics.mean(window)  
median = statistics.median(window)  
variance = statistics.variance(window)  
stdev = statistics.stdev(window)  
maximum = max(window)  
minimum = min(window)  
data_range = maximum - minimum
```

Interpretation: Calculates all seven statistical measures for the current window.

Step 6: Display Results

```
print(mean)  
print(median)  
print(variance)  
print(stdev)  
print(maximum)  
print(minimum)  
print(data_range)
```

Interpretation: Displays the statistics of the current window before the window moves forward.

Complete Program

```
import statistics  
import random  
  
stream = [random.randint(10, 100) for i in range(20)]  
  
window_size = 5  
print("Window size:", window_size)  
  
for i in range(len(stream) - window_size + 1):  
  
    window = stream[i : i + window_size]
```

```

mean = statistics.mean(window)
median = statistics.median(window)
variance = statistics.variance(window)
stdev = statistics.stdev(window)
maximum = max(window)
minimum = min(window)
data_range = maximum - minimum

print("\nWindow", i + 1, ":", window)
print("Mean      :", mean)
print("Median    :", median)
print("Variance  :", variance)
print("Std Dev   :", stdev)
print("Max       :", maximum)
print("Min       :", minimum)
print("Range    :", data_range)

```

Sample Input

```

stream = [27, 82, 18, 42, 25, 73, 67, 70, 93, 58,
          36, 22, 72, 13, 59, 65, 87, 10, 99, 67]

```

```

window_size = 5

```

Expected Output

```

Window size: 5

```

```

Window 1 : [27, 82, 18, 42, 25]

```

```

Mean      : 38.8

```

```

Median    : 27

```

```

Variance  : 659.7

```

```

Std Dev   : 25.68462575160479

```

```

Max       : 82

```

```

Min       : 18

```

```

Range     : 64

```

```

Window 2 : [82, 18, 42, 25, 73]

```

```

Mean      : 48

```

```

Median    : 42

```

```

Variance  : 811.5

```

```

Std Dev   : 28.48683906648823

```

```

Max       : 82

```

```

Min       : 18

```

```

Range     : 64

```

```

...

```

```

Window 16 : [65, 87, 10, 99, 67]

```

A total of 16 windows are displayed. The exact values change on every run because the stream is generated randomly.

Interpretation

- The stream contains 20 randomly generated values between 10 and 100.
- The window size is fixed at 5.
- The window moves forward by one position as new data enters.

- Only the current window is used for the calculation, so memory usage stays constant.
- Statistics are recalculated for every window.
- A total of 16 windows are created, since $20 - 5 + 1 = 16$.
- The changing mean and range across the windows show how the stream behaves over time.

Result

The streaming data was successfully processed using the sliding window technique. Statistical measures such as mean, median, variance, standard deviation, maximum, minimum and range were calculated for each window.